Recurrent Latent-Space Reasoning: Watching a Model Think Without Words

DataForge 2026 × Pathway × rime — Pathway Track Submission

The one-sentence claim

A model can improve an answer by repeatedly transforming a fixed-size hidden state a variable number of times — without ever converting an intermediate step into a word or token.

This is a claim about *mechanism*, not vibes, and it is written so it can be proven false. If it were wrong, giving a model more "thinking passes" over the same hidden state would do nothing: accuracy would sit flat no matter how many iterations you allowed. Our artifact is built so a learner can run exactly that test in about ten seconds.

Why this idea matters right now

Most people's mental model of "AI reasoning" is chain-of-thought: the model writes out intermediate steps in natural language — "first I'll check X, then Y" — and each step costs real output tokens. That approach works, but it is expensive, and it forces every intermediate thought to be serialized into words even when the underlying computation isn't naturally verbal.

An alternative line of research asks: what if the model reasoned by repeatedly transforming a *hidden state*, the way a person might turn a problem over in their head before speaking? No tokens are spent on intermediate steps; a fixed-size vector (or belief distribution) is refined, refined again, and only decoded once, at the end. Hao et al. (2024) first demonstrated this for autoregressive LLMs by training a model to reason in continuous latent space instead of words. Geiping et al. (2025) showed a recurrent-depth architecture that scales test-time compute by looping a shared block over a latent state rather than growing the model or the token count. Saunshi et al. (2025) gave this idea theoretical footing, showing looped transformers can simulate bounded chain-of-thought computation without adding parameters per step.

Pathway's BDH-CQ system is a recent, concrete instance of this same idea, which is what makes it a natural anchor for an explainer built around this concept.

What the artifact actually does

The interactive page (`index.html`) is a small, real, running model — not an animation. It works like this:

1. **Two demonstrations.** The learner is shown two input-grid → output-grid pairs. Somewhere in that pair is a hidden rule: the grid has been shifted by some (dx, dy) offset and its four colors have been permuted by some fixed swap.
2. **A hypothesis space, not a guess.** The model never invents a rule from scratch. It holds a fixed set of 54 candidate rules — 9 possible offsets times 6 possible color permutations — and scores every one of them against how well it reproduces the two demonstrations.
3. **A reasoning-effort dial.** Buttons (0 / LOW / MEDIUM / HIGH, corresponding to 0, 1, 4, and 14 iterations) or an exact slider (0–20) control how many times the model's belief over those 54 hypotheses gets refined. Evidence accumulates linearly with iteration count, then passes through a softmax, so more iterations genuinely sharpen the belief toward whichever hypothesis the demonstrations best support.
4. **A visible belief state.** The top candidate hypotheses are drawn as bars that move live as the dial changes. At no point does the model write "I think the rule is a shift of (1, 1)" — the belief only ever exists as numbers.
5. **A held-out query.** The model applies its current best hypothesis to a new grid it hasn't seen an answer for, and that answer is placed directly beside the true answer, so the gap (or lack of one) is visible rather than asserted.

Run the falsification test yourself: set the dial to 0 iterations and note the accuracy (it's the same as guessing — 25%, chance level for four colors, because with no evidence integrated the belief is uniform over all 54 hypotheses). Then jump to HIGH. In the included worked example, one iteration is already enough for the demonstration evidence to single out the correct rule, after which accuracy saturates at 100% and the belief on the correct hypothesis keeps sharpening (its probability keeps climbing toward 100%, and the residual error keeps shrinking on a log scale) for many further iterations, even though the answer itself stopped changing. That distinction — the model growing more *confident*, not just more *accurate*, as it "thinks" longer — is itself a lesson embedded in the interaction, and it's a real, reproducible property of the underlying math (see the companion note, *Recurrent Latent-Space Reasoning: Mathematical Formulation*, for the exact equations and a re-run of this exact computation across all 21 effort levels).

One panel on the page is explicitly *not* live: a static, hand-written illustration of what a token-based reasoner's chain-of-thought trace would look like on the same puzzle, labeled in-page as illustrative so it is never mistaken for the model's actual output.

Where BDH and BDH-CQ fit

The toy model above is a teaching device, not a reproduction of any Pathway system. But its mechanism — a fixed-size state, refined over several passes, never decoded to words along the way — is exactly the shape of the mechanism BDH and BDH-CQ implement at a much larger scale, which is the reason this concept was chosen for the BDH module.

BDH (Dragon Hatchling) gives a sequence model a persistent, synapse-level state. Instead of attention being computed by comparing every pair of tokens directly, attention emerges from local neuron-to-neuron interactions on a sparse, scale-free graph, updated through Hebbian-style rules where recently co-active connections strengthen. The practical effect is a fixed-size, continuously-updatable state instead of a growing key-value cache.

BDH-CQ is what happens when that state is used for two distinct jobs at once. Demonstrations continuously update a *contextual memory* — this plays the same role as our two demo grids, adapting the system's behavior without touching its trained weights. Separately, a *latent workspace* is transformed repeatedly to solve the query — this is precisely the iteration-count dial in our demo, generalized to a much higher-dimensional space. Critically, none of this involves a backward pass or a parameter update at inference time; the adaptation lives entirely in recurrent state.

Reported figures from the primary paper (Engdahl et al., 2026, arXiv:2608.09888): a 150M-parameter BDH-CQ configuration reaches 29.5% pass@2 / 24.25% pass@1 on the public ARC-AGI-1 evaluation set, at a computed inference cost of roughly $0.00070 per task. Increasing reasoning effort tracks accuracy in the same direction our demo dial does: LOW effort scores 21%, MEDIUM 27%, HIGH 29.5% pass@2. These are developer-reported figures, shown for reference and comparison — never presented as live output of our toy model, and our toy model does not use BDH-CQ's weights, code, or unpublished implementation details.

What this doesn't prove

More latent iterations is not the same as a bigger or smarter model. Turning the dial up in the demo doesn't add parameters or knowledge — it only gives the existing belief state more passes to integrate the same fixed evidence. The same caveat applies to BDH-CQ itself: its recurrent contextual-memory update is not a weight update. Nothing about the model's trained parameters changes when it adapts to new demonstrations, which is a genuinely different mechanism from architectures that fine-tune or backpropagate at test time.

A real limitation, stated plainly

BDH-CQ's own controlled studies show robust extrapolation on tasks like boundary propagation, copying, and dense mappings — but performance drops sharply on long ordering sequences, deep nesting, conditional rule selection, and composing multiple transformations together (one reported composition — color-swap combined with relocation — was solved 0 times out of 72 attempts). Our toy demo deliberately uses a small, disambiguated hypothesis space so that it always converges cleanly; the real system does not always converge this cleanly, and that gap is worth sitting with rather than smoothing over.

A note on evidence quality

One number worth being precise about: the BDH-CQ paper's public ARC-AGI-1 result includes a black-box audit conducted by co-authors of the BDH-CQ paper itself, affiliated with partner institutions. It is a genuine audit without access to model weights, but it is not a fully external third-party reproduction, and we label it that way rather than as an unqualified "independent evaluation" — the distinction matters because a result checked by an uninvolved third party carries different evidential weight than one checked by people connected to the original team.

Try it yourself

Open the artifact, leave the dial at 0, and watch the model guess. Then move it to HIGH and watch the belief bars sharpen and the query answer land on the true grid. That's the whole claim, made visible.

Primary sources

- Hao, S., Sukhbaatar, S., Su, D., et al. (2024). *Training Large Language Models to Reason in a Continuous Latent Space.* arXiv:2412.06769.
- Geiping, J., McLeish, S., Jain, N., et al. (2025). *Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach.* arXiv:2502.05171.
- Saunshi, N., Dikkala, N., Li, Z., Kumar, S., Reddi, S. J. (2025). *Reasoning with Latent Thoughts: On the Power of Looped Transformers.* arXiv:2502.17416.
- Kosowski, A., Uznański, P., Chorowski, J., Stamirowska, Z., Bartoszkiewicz, M. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain.* arXiv:2509.26507.
- Engdahl, B., Kosowski, A., Chorowski, J., et al. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning.* arXiv:2608.09888.